

罗布莎豆荚状铬铁矿综合地球物理勘探

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摘要: 豆荚状铬铁矿是铬铁矿的主要类型之一, 中国和世界各地的铬铁矿大多数是豆荚状铬铁矿。由于普遍发育在碰撞、挤压的构造环境, 尖灭再现严重, 豆荚状铬铁矿是一个全球性的勘探难题。冈底斯东段罗布莎是我国规模最大的铬铁矿产地, 在维系我国的铬铁矿对外依存度和战略安全方面发挥着重要作用。但矿山面临严重的资源接替危机, 先期的地球物理工作没有重大勘探发现, 矿山多年没有勘探突破。由于铬铁矿具有显著的密度、磁化率特征而电阻率特征不明显, 传统的观点认为铬铁矿应该以重磁勘探为主, 电磁法难以发挥作用。我们在罗布莎开展重力、音频大地电磁法 (AMT) 和激电方法等铬铁矿找矿方法技术研究取得找矿突破, 2012~2014 年, 部署的物探异常验证孔 7 个见矿, 其中 ZKWT02 孔见矿 49.18 米, 发现了我国规模最大的铬铁矿, 有效缓解了矿山的资源危机。研究结果表明, 罗布莎深部依然存在较好的找矿前景, 物探方法特别是电法能在铬铁矿深部找矿中发挥较好的作用。

关键词: 豆荚状铬铁矿, 综合物探, AMT, IP, 罗布莎

1 前言

铬铁矿是一种重要工业矿产资源, 中国铬铁矿资源极度稀缺, 铬铁矿进口逐年增加, 对外依存长期处于高位, 形势十分紧迫(王希斌等, 2010)。罗布莎是我国最大的铬铁矿生产基地, 矿山存在接替资源短缺的问题。目前地面铬铁矿已基本开采殆尽, 深部(200 米以下)铬铁矿已成为新的勘探开发目标。传统的观点认为铬铁矿应该以重磁勘探为主, 电磁法难以发挥作用。铬铁矿勘探已有近百年的历史, 国内外普遍采用地球物理勘探技术寻找隐伏铬铁矿。概括起来, 物探技术在铬铁矿勘探中主要发挥两方面的作用 (Fraseri, 2009): (1) 直接找矿; (2) 寻找控矿构造。自上世纪四十年代开始至今, 古巴、希腊、苏联、土耳其、阿尔巴尼亚、印度、菲律宾等国家先后开展了铬铁矿地球物理勘探工作并有成功的实例, 阿尔巴尼亚、古巴、印度国家开展了新一轮的地球物理勘探工作。但总体上, 地球物理方法勘探铬铁的效果并不明显 (Fraseri, 2009)。姜枚等 (2015) 选择重、磁、大地电磁法和地震等的剖面探测, 在罗布莎、泽当、东坡岩体进行综合地球物理研究工作, 认为综合各种地球物理场的特征认为在磁异常分布区内具有局部重力异常、低阻体异常较多的地段是岩体最有利于进一步详细探测找矿的部位。

本文以罗布莎豆荚状铬铁矿勘探为研究目标, 通过重力、AMT 和精密激电等

物探方法组合,研究了罗布莎岩体的电磁学特征和铬铁矿含矿构造的综合异常特征,在此基础上部署了勘探异常验证钻孔,并取得了重大勘探突破。

2 地质背景

罗布莎铬铁矿位于雅鲁藏布蛇绿岩带的东部,是一种阿尔卑斯型的铬铁矿。罗布莎蛇绿岩主要由三个岩相单元组成:变质地幔橄榄岩、基性—超基性堆晶岩以及辉长岩、辉绿岩岩脉。超过 93%的岩体由变质地幔橄榄岩组成,主要变质矿物有蛇纹石和磁铁矿。地幔橄榄岩主要包含纯橄岩、斜辉辉橄岩和辉石橄榄岩(图 1)。罗布莎铬铁矿床有以下特征:(1)矿体成群出现,分段集中。(2)矿床可以划分为一个含矿构造带(即中央含矿带),两个边缘矿带和一个矿化带(张浩勇等, 1996; 王希斌等, 2010)。(3)矿体主要赋存在斜辉辉橄岩中。(4)矿体主要受东西向断裂带控制,其他几组虽控矿但规模不大,仅具联合控矿的作用。(5)对罗布莎豆荚状铬铁矿具体成因机制的看法仍有较大的分歧。主要有“地幔演化—动力成矿”、“岩石-熔体反应”和“深部成因”模式三种模式 (Zhou et al, 2005)。

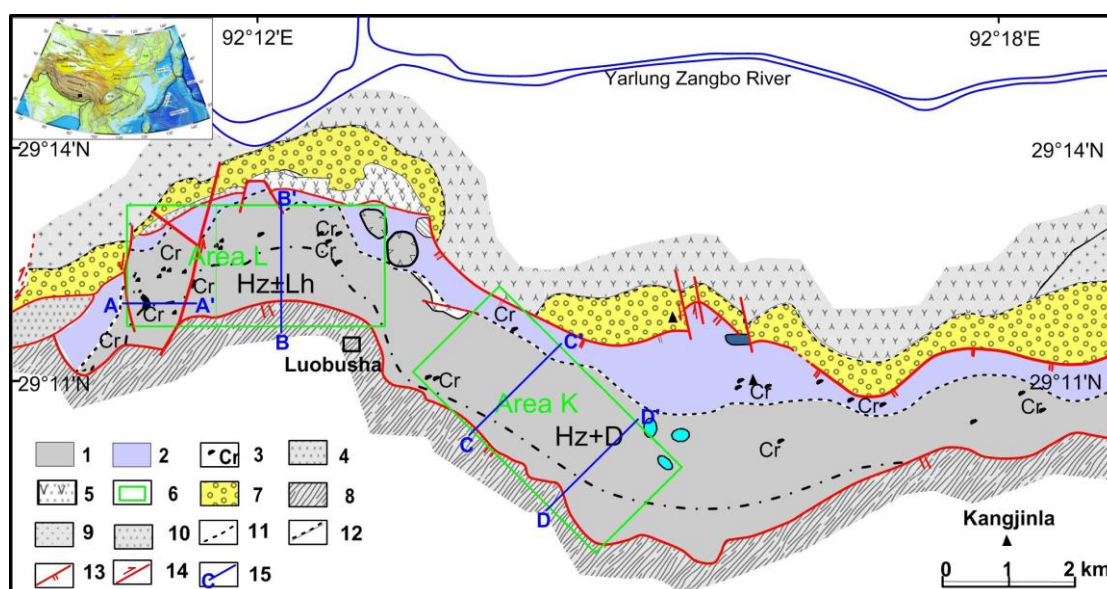


图 1 罗布莎蛇绿岩地质简图(修改自李金阳等, 2012)

1-方辉橄榄岩(含纯橄岩和二辉橄榄岩); 2-纯橄岩; 3-铬铁矿体; 4-辉长岩为主的堆晶岩; 5-辉长岩异剥橄榄岩和辉石岩组成的堆晶岩; 6-重点研究区; 7-第三系罗布莎群; 8-上三叠统地层; 9-石英闪长岩, 石英二长岩; 10-黑云母花岗岩; 11-岩相界限; 12-不整合接触界限; 13-逆断层; 14- (走滑) 断层; 15-典型剖面位置

3 数据分析

野外采用了重力、音频大地电磁测深法 (AMT) 和激电法。AMT 观测了超过数千个测点, 勘探范围为 10-km², 主要部署在香卡山和罗布莎矿区(图 1)。罗布莎西段测线方位为 90°, 东段为 0°, 香卡山矿区 42°。数据采集采用了凤凰公司的 MTU-5A 和湖南强军公司自主开发的 PMT-2 音频大地电磁仪, 点距 20m, 电极距一般为 20 米。使用凤凰公司数据处理软件包进行现场数据处理。地形和静态效应

采用电磁阵列剖面滤波法 (EMAP) 进行过滤, 数据反演非线性共轭梯度法两种方法。

激电采用湖南强军公司研发生产的无线组网多通道激电仪系统。所采集参数主要为视电阻率和相对相位等, 探测深度可达 1000 米。激电测深法的工作原理和传统激电法基本相同, 但它采用基于无线传感器网络的阵列观测技术, 一次供电多点观测。

重力采用 CG-5 观测, 点距 20 米, 数据反演采用三维重磁物性层离相对密度反演法。

4 结果与讨论

图 2 为香卡山矿区 AMT 反演电阻率高阻(a)和低阻(b)空间分布图。从图中可以看出, 香卡山矿区高低阻异常主要呈条带状相间分布。东南方向发育连续分布的高阻条带, 近北东南西走向, 长度超高 3000 米。在它的北东方向近似平行发育了不连续分布的高阻条带, 近北东南西走向, 长度超高 4000 米。高阻区的电阻率大于 $1000\Omega\cdot m$ 岩体主要分布两个条带。低阻区和高阻区相间分布, 主要发育三个低阻条带, 北东部低阻条带、中部低阻条带和南西部低阻条带, 其中中部低阻条带发育最完整。中部低阻条带在北西部为近东西走向, 在南东段扭转为近北东南西走向。通过已知铬铁矿的岩石电磁学剖面特征对比分析认为, 铬铁矿主要分布在高低阻异常过渡带上。依据和已知矿的对比分析和物探异常验证孔的钻探结果, 基于电磁特征的异常模式, 综合岩石电学特征, 我们提出三种找矿模式: 第一种为高阻接触带模式。这种模式的矿体规模相对较大。第二种为低阻圈闭模式。第三种为低阻裂隙模式, 这种模式矿体规模较小, 成功率较低。图 3 为综合物探异常预测模式图, 可以看出, 重力相对密度、激电相对相位和 AMT 电阻率异常之间存在较好的对应关系, 三种异常的叠合部位解释为铬铁矿的含矿有利区。2012 年到 2014 年间, 由矿业公司和国家老矿山项目部分投入对上述异常模式的钻孔分别进行了验证。第一种模式布设验证孔 2 个, 钻孔验证见矿 2 个。其中验证孔(02)钻遇厚达 49.18m 的铬铁矿(图 3b), 大部分矿体品位超过 50%。第二种模式布设验证孔 4 个, 钻孔验证见矿 3 个, 最大见矿厚度 7.25m, 最小 0.46m。第三种模式布设验证孔 3 个, 钻孔验证见矿 2 个, 最大见矿厚度 2.75 米, 最小 0.30m。

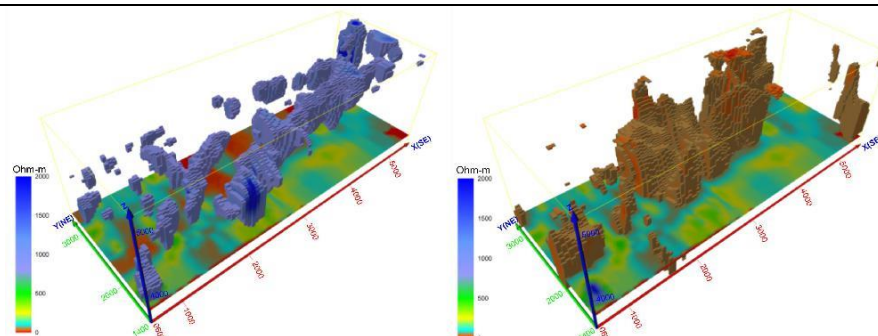


图 2 Area K 矿区三维电阻率成像图 (左:高阻成像, 右: 低阻成像)

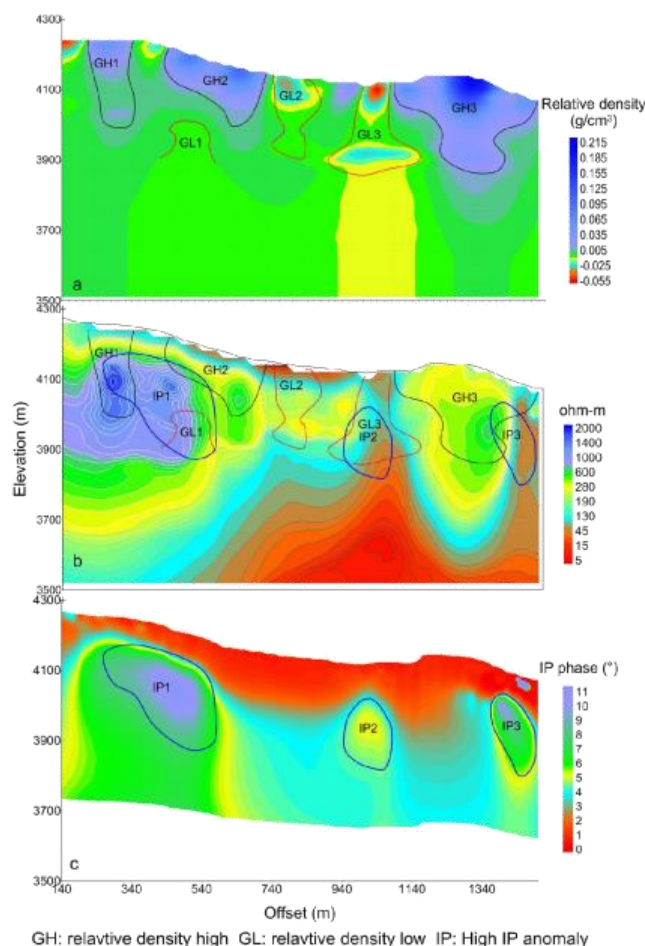


图 3 重力、AMT 和激电综合的找矿模型图

5 结论

在西藏南部隐伏深埋豆荚状铬铁矿中开展了综合地球物理勘探。结果表明：AMT 异常和铬铁矿含矿构造之间存在一定的对应关系。依据和已知矿的对比分析和物探异常验证孔的钻探结果，基于电磁特征的异常模式，综合岩石电学特征，我们提出高阻过渡带、低阻圈闭和低阻裂隙三种找矿模式。铬铁矿在整体上和重力和激电都异常存在一定的对应关系，铬铁矿和含矿围岩组成的含矿构造(或含矿单元)内部常常会存在重磁异常平衡，应用重磁勘探识别这种隐伏的含矿单元难度较大。

应用综合地球物理勘探能够有效地降低勘探风险，并已取得重大勘探突破。

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Comprehensive exploration for podiform chromites in the Luobusa ophiolite

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ABSTRACT: The exploration of podiform chromites in the Luobusa ophiolite of southern Tibet has proved difficult in recent decades because most known deposits pinch out and then reappear in the same direction. A number of ground-based geophysical approaches, including gravity, magnetic, and controlled source audio-frequency magnetotelluric methods have been applied to explore for these chromite deposits but have mostly failed to delineate prospective areas. We present a successful podiform chromite exploration case history that is based on comprehensive geophysical exploration including gravity, audio magnetotellurics (AMT) and induced polarization (IP). AMT results show correlation to rock faces in terms of fresh harzburgite, altered harzburgite, and dunite, and can thus be connected to concealed deposits. We therefore propose three resistivity anomaly models for chromite exploration. Based on the result of AMT, gravity and IP, we delineate a number of prospective regions containing exploitable deposits within the Luobusa ophiolite. Seven of the nine verified boreholes discussed in this paper intersected with chromite deposits; one comprises the largest and highest-grade chromite deposit in China up to date.

Keywords: Podiform chromite, comprehensive exploration, audio magnetotelluric, Luobusa ophiolite, Tibet

1. INTRODUCTION

The Luobusa ophiolite contains the largest reservoir of chromite in China, encompassing almost 90% of total national production (Wang et al., 2010). Systematic exploration and studies have been carried out on the Luobusa ophiolite since the initial discovery of chromite in 1956 (Zhang et al., 1996). In spite of research progress on the origins and basic geology of podiform chromites, the exploration for concealed deposits has proved problematic (Fraseri, 2009). Gravity and magnetic methods have been widely applied for chromite exploration. A limited amount of geophysical exploration has been conducted over the last century in the Luobusa ophiolite and its vicinity, but the results were unconvincing to geologists as only a few ore deposits has been discovered by drilling based on the geophysical interpretations. Exploration for chromite has become a key challenge in Luobusa (Wang et al., 2010). In order to explore for and evaluate concealed and potential chromite deposits within the Luobusa ophiolite, we conducted a comprehensive exploration project financed by mining companies between 2012 and 2013. We applied detailed gravity, magnetic, AMT, and IP methods using spread spectrum technology over a 10 km² area within the ophiolite region to search for chromite. AMT and IP were the major geophysical exploration methods applied in this project; we identified several favorable areas and located a number of economic chromite deposits via exploratory drill cores. One drill core that was based on AMT result with a non-linear conjugate gradient inversion (NLCG) interpretation resulted in the discovery of four layers of chromite with a total thickness of 49.18 m, the largest deposit found so far in China.

2. GEOLOGICAL SETTING

The chromite deposits within the Luobusa ophiolite are associated with Alpine-type peridotites, one of a number of such assemblages within the Yarlung Zangbo ophiolite belt (Figure 1). The Luobusa ophiolite is about 40 km in length and ranges in width between 0.7 km and 4 km. Based on available outcrop observations and drill core results, the major lithotypes within the Luobusa ophiolite are identified as harzburgite and dunite (Figure 1). The contact between these two lithotypes is characterized as the central ore bearing zone, and most chromite appears to occur within, or close to this zone. The major elemental constituents of chromites in both podiform chromites and mantle peridotites in the Luobusa ophiolite were reported by Zhang et al. (1996) and Zhou et al. (2005). The transition band between these two major assemblages plays an important role in high-grade chromite exploration and its burial depth ranges from several tens of meters to several hundreds of meters. The presence of high-grade deposits is also controlled by faults that generally strike from west to east. Most of the known chromite deposits are podiform and pinch out and reappear in the same direction, creating issues for geological and geophysical exploration (Zhang et al., 1996).

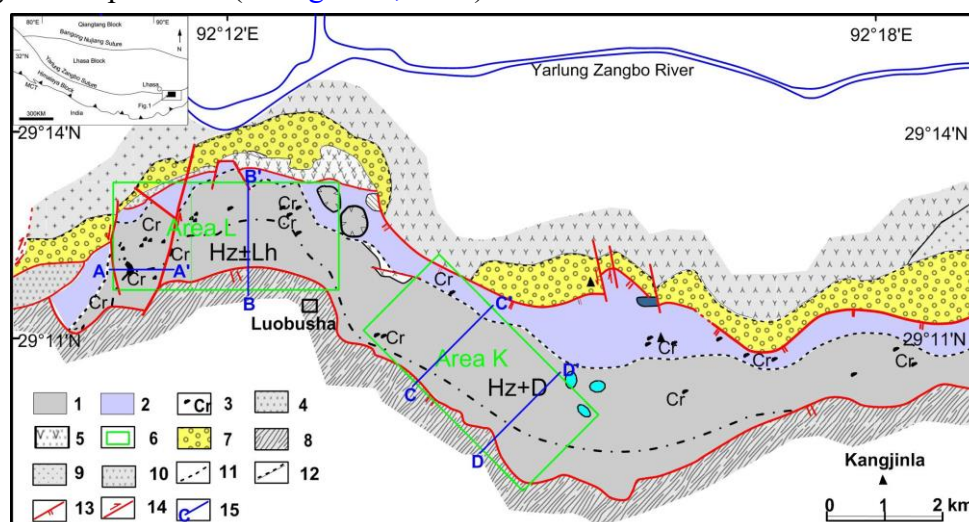


Figure 1. Schematic geological map of the Luobusa Ophiolite (He et al., 2014, modified after Li et al., 2012). The black filled rectangle at the top left corner shows the location of the working area in Tibet.

3. METHODS

We conducted Gravity, AMT and IP in Area K and L in the the Luobusa ophiolite. AMT was the dominant method we utilized for comprehensive podiform chromite exploration within the Luobusa ophiolite. AMT surveyed with line spacing of 80 m or 40 m, and the normal station spacing was 20 m. Survey line directions were laid out across the strike of the ophiolite belt with a direction of 0°, 90° or 42°. We used the software package SSMT2000 (Phoenix Geophysics) for onsite data processing. We utilized electromagnetic array profiling (EMAP) filtering as a post-processing step to correct topographic and static shifts in data along the survey line direction (XY) before inversion or conversion. Non-linear conjugate gradient inversion (NLGG) was applied for AMT data inversion. IP data were collected using the instruments developed by Champion

Geophysics with a grid of 20*40 m. Gravity data was acquired by CG-5, with a grid of 20*40 m or 20*80 m. 3D inversion of relative density was used for data conversion.

4. RESULTS AND DISCUSSION

3D resistivity image results for Area K are shown in Figure 2. Figure 2a illustrates face-render results from rock mass that have resistivities higher than 800Ω.m. Figure 2b illustrates those that have resistivities lower than 50Ω.m. From the surface to a depth of 1,500 m, the rock mass in Area K can be divided generally into a higher resistivity upper domain and a lower resistivity basal domain. A continuous and thin high-resistivity block that yields values greater than 800Ω.m trends from northwest to southwest across Area K (Figure 2b). Another similar and discontinuous thin high resistivity block with comparable properties trends from northwest to southeast and both are characterized by burial depths that range between 50 m and 700 m. Rocks characterized by lower resistivity values are interpreted as mainly altered dunite. The rest part of the ophiolite is characterized by resistivities that range from 50Ω.m to 800Ω.m intermediate between high resistivity blocks and the low resistivity volume. They are reflections of partially serpentinized harzburgite and dunite. The majority of chromite deposits occur within this transitional zone.

Figure 3 shows the comprehensive result of AMT, gravity and IP in same section in the middle area L. Figure 3a show the relative density result extracted from 3-D gravity inversion result. Figure 3b is AMT result overlaid the density and IP anomalies, and Figure 3c is inverted phase of data of IP based on spread spectrum technology. We interpreted three high relative density anomalies (GH1-3) and three high relative density anomalies (GL1-3) in the relative density section. Compare to AMT result, three GH and two GL anomalies have the same horizontal position to the AMT high resistivity anomalies and low resistivity anomalies, but the center of GH anomalies is shallower than AMT high resistivity anomalies. Three IP anomalies were interpreted in the same section, two of them (IP 2 and IP 3) lies into the transition zone from high resistivity to low resistivity of AMT section (Figure 3b).

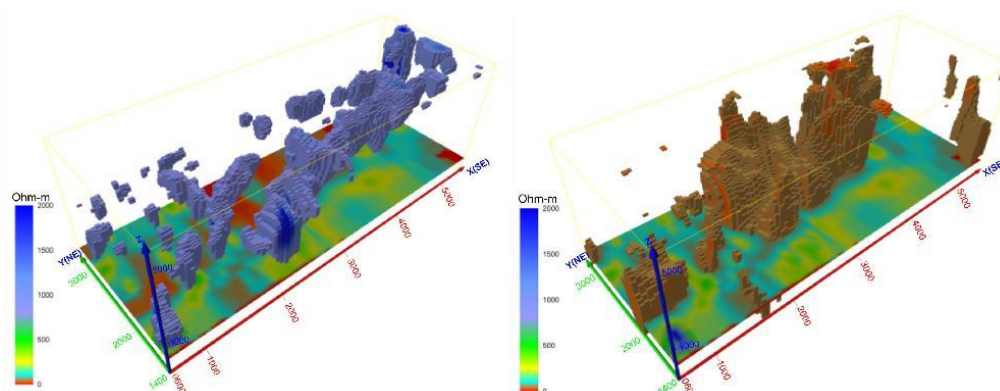


Figure 2. Three dimensional resistivity view of Area K at Luobusa ophiolite based on AMT inversion data.

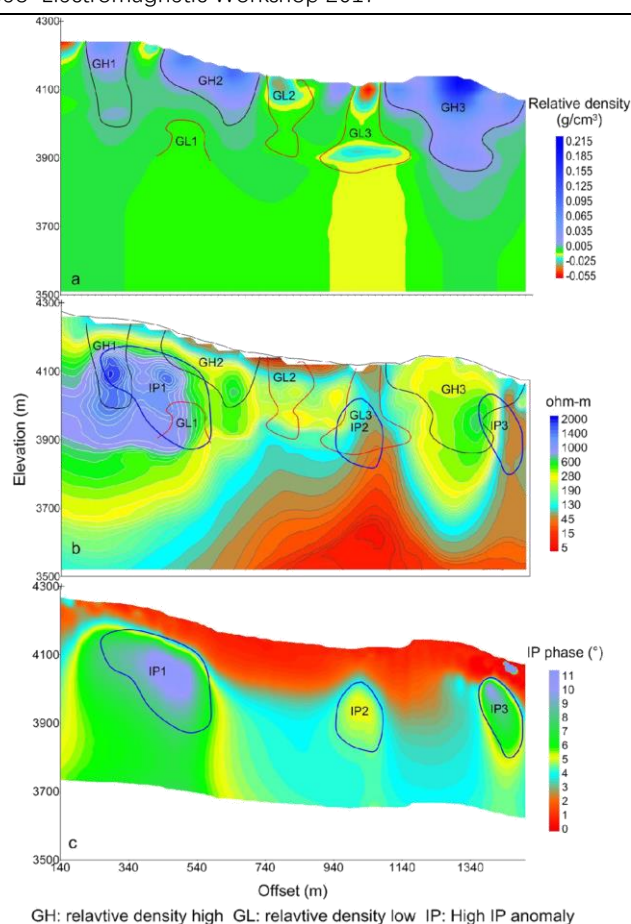


Figure 3. The Comprehensive exploration result of Luobusa podiform chromite

The AMT result reveals the relationship between a known chromite deposit and the resistivity cross section. A known chromite deposit occurs within the transition zone between higher resistivity (fresh harzburgite) and lower resistivity domains (serpentinized harzburgite and dunite). On this basis, we propose the use of a ‘transition zone’ model (T model) for chromite exploration. Drill cores within this area intersected a new chromite deposit that has a maximum thickness of 49.18 m, including two thin sheets of rock with a total thickness of about 3 m. We also present two additional chromite exploration models based on the resistivity results discussed in this study. The first of these is termed as the ‘lower resistivity fracture’ model (F model) where a deposit is characterized by low resistivity and occurs in an ophiolite fracture, especially relatively shallow ones. The second of our two additional models is referred to ‘lower resistivity entrapment’ model (E model) where a deposit is characterized by low resistivity, enclosed by high resistivity host rocks.

Between 2012 and 2014, nine proposed boreholes were drilled to verify chromite the geophysical exploration results. Of these, two are consistent with the T model, and both intersected high-grade deposits. One of them, Borehole 02, led to the discovery of the largest and highest-grade Chinese chromite deposit to date. Five additional boreholes were drilled on the basis of the E model, and four intersected chromite deposits with thicknesses ranging between 0.6 m and 5.6 m, and the fifth did not intersect any mineralization. Finally, two boreholes were drilled on the basis of the F model, and one of them intersected a thin deposit.

5. CONCLUSION

We present a case study applying a comprehensive exploration for deeply buried podiform chromite within the Luobusa ophiolite of southern Tibet. We conducted AMT, gravity and IP along more than 100 sections across the whole rock mass. Typical sections

of the Luobusa ophiolite combined with a 3D view of our AMT results show that this rock mass can generally be divided into three domains, a shallow low resistivity region that is indicative of loose Quaternary system deposits, a middle domain characterized by highly variable resistivity due to the presence of fresh harzburgite, altered harzburgite and dunite, and a deep very low resistivity domain interpreted as altered dunite containing high water content. We propose three resistivity models for future chromite exploration. Our proposed models are effective and the largest chromite deposit in China up to date has been uncovered by boreholes based on our approach. Our results have led to the discovery of new favorable areas for chromite exploration within the Luobusa ophiolite. Data from the comprehensive surveys also significantly refine our understanding of the structure and distribution of rock faces within the Luobusa ophiolite.

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